

Homework 6

Sonia Kangaju & Abbey Skinner

2026-04-20

Model Specification (25 points)

1. Outcome and link function (10 points): Identify the binary, count, or discrete outcome you are analyzing. Specify an appropriate link function and variance function for your outcome. Justify your choices.

Our outcome is systolic blood pressure, which is positive and continuous response. We model the marginal mean using Gamma distribution and a log link specified as $(g(\mu_{ij}) = \log \mu_{ij} = X_{ij}\beta)$ where $E(Y_{ij}) = \mu_{ij} = e^{X_{ij}\beta}$. We assume a Gamma variance structure, $Var(Y_{ij}) = \phi\mu_{ij}^2$. We chose the Gamma distribution because SBP is continuous and strictly positive, and the log link ensures positive fitted values while allowing multiplicative interpretation of covariate effects.

2. Mean model (15 points): Write out your marginal mean model. Identify the covariates you are including and describe what each parameter represents. Remember that parameters in marginal models have population-averaged interpretations.

Let Y_{ij} be the systolic blood pressure for subject i at the j th visit. Note that marginal models handle time continuously, where it enters as a covariate in X_{ij} . Additionally, since marginal models do not include a random effect, we will only have the population mean trajectory (no subject specific mean trajectory). We can write the marginal mean model as:

$$E(Y_{ij}) = \mu_{ij} = \exp(\beta_0 + \beta_1 x_{i,YEAR} + \beta_2 x_{i,SEX})$$

- β_0 denotes the baseline log mean SBP for males
- β_1 denotes the log multiplicative change in mean SBP per year
- β_2 denotes the log ratio in mean SBP between females and males at any fixed point in time
- $x_{i,YEAR}$ denotes the number of years since baseline for the i th subject
- $x_{i,SEX}$ is a binary indicator for the sex of the i th subject

Working Correlation Structure (25 points)

1. Fitting the model (10 points): Fit your marginal model using GEE with at least two different working correlation structures (e.g., independence, exchangeable, AR(1), unstructured). Report the parameter estimates from each.

We will fit our model with three different working correlation structures: exchangeable (compound symmetry), AR(1), and unstructured. Table 1 below gives the parameter estimates, standard errors, Wald test statistics, p-values, and the QIC for each model.

Table 1: Comparison of GEE estimates across working correlation structures

Working Correlation	Parameter	Estimate	SE	Wald Statistic	p-value	QIC
Compound Symmetry	(Intercept)	4.862	0.003	2034214.11	<0.0001	19970
Compound Symmetry	YEAR	0.006	0.000	921.42	<0.0001	19970
Compound Symmetry	SEX	0.013	0.005	7.66	0.0056	19970
AR(1)	(Intercept)	4.861	0.003	2037638.16	<0.0001	19991
AR(1)	YEAR	0.006	0.000	919.59	<0.0001	19991
AR(1)	SEX	0.013	0.005	7.33	0.0068	19991
Unstructured	(Intercept)	4.862	0.003	2047439.73	<0.0001	19976
Unstructured	YEAR	0.006	0.000	919.63	<0.0001	19976
Unstructured	SEX	0.013	0.005	7.61	0.0058	19976

2. Comparing structures (15 points): Compare the results across working correlation structures. How sensitive are your parameter estimates to the choice of working correlation? Select a working correlation structure and justify your choice.

Regardless of the covariance structure, all terms are found to be significant in the model and all estimates are generally very similar to each other. This tells us that our parameter estimates are not very sensitive to the choice of working correlation. This aligns with the fact that parameter estimates are consistent even when the covariance structure is misspecified. When looking at the QIC for each model, we yield the smallest QIC under compound symmetry covariance structure (exchangeable). Because of this, we will choose compound symmetry as our working covariance structure.

Variance Estimation (20 points)

1. Comparison (10 points): For your selected working correlation structure, compare the model-based and sandwich (robust) standard errors. Create a table showing both sets of standard errors.

Table 2: Comparison of model based and robust standard errors for exchangeable covariance structure

Parameter	Estimate	Model Based SE	Robust SE
(Intercept)	4.862	0.004	0.003
YEAR	0.006	0.000	0.000
SEX	0.013	0.005	0.005

2. Interpretation (10 points): Discuss what the differences (or similarities) between these standard errors suggest. Given your sample size and the robustness properties of GEE, which variance estimator do you consider more appropriate for your analysis, and why?

The model-based standard errors are similar to the robust standard errors for all three estimates. The similarity between these errors suggests that the selected working covariance structure is a reasonable specification. Despite this, we will still use the robust standard errors to be safer as the model based standard errors rely on the covariance structure being correctly specified.

Hypothesis Testing and Interpretation (20 points)

1. Hypothesis test (10 points): State and test a hypothesis relevant to your research question. Report the test statistic, degrees of freedom, p-value, and interpret the result.

We will test to see if there is a sex effect. Our hypotheses are $H_0 : \beta_2 = 0$ vs $H_1 : \beta_2 \neq 0$. We will look at the compound symmetry, sex row of Table 1 to find the relevant statistics to report for this test. This test has 1 degree of freedom and yields a test statistic of 7.66 and a p-value of 0.0056. Since the p-value is < 0.05 , we will reject H_0 . We claim that there is a significant multiplicative difference in mean SBP between males and females after adjusting for the number of years since baseline.

2. Interpretation of effects (10 points): Interpret your key parameter estimates in the context of your research question. Be specific about the population-averaged interpretation (e.g., for a binary outcome with logit link, interpret in terms of odds ratios at the population level).

The compound symmetry section of Table 1 presents the parameter estimates discussed below:

- $\beta_0 = 4.862$ represents the population-averaged log mean SBP at baseline for males. Equivalently we can state that mean SBP at baseline for males is $\exp(4.862) = 129.28$.
- $\beta_1 = 0.006$ indicates that, on average, SBP increases by a factor of $\exp(0.006) = 1.006$ per year at the population level.
- $\beta_2 = 0.013$ indicates that on average, females have a mean SBP that is $\exp(0.013) = 1.013$ times that of males, after adjusting for the number of years since baseline.

Recall that we have a continuous outcome and used the log link function in our model.

Summary (10 points)

Summarize your findings as you would for a collaborator. Address:

- What model did you fit and what working correlation structure did you use?
- What are the key findings regarding your research question?
- What are the limitations of the marginal model approach for your data?

We fit a marginal model for systolic blood pressure using generalized estimating equations with a Gamma distribution and a log link, including time and sex as covariates. We specified an exchangeable (compound symmetry) working correlation structure to account for within-subject correlation across repeated measurements.

We found that both time and sex are significantly associated with systolic blood pressure at the population level. Specifically, at the population level, we found that SBP increases multiplicatively over time and that females have higher SBP than males on average at any given time point.

One limitation of the marginal model is that we are not able to calculate subject specific mean trajectories, since a random effect is not included in our model. Because of this, we are not able to capture individual level variability in baseline SBP or its change over time. Lastly, the model based standard errors require that the working covariance structure is correctly specified; however, the robust standard errors provide an alternative for when the covariance structure may be misspecified.